

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer’s advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower semantic store consolidated by target- k KMeans. Soft O2 binds session siblings by attenuated score inheritance (βs) rather than hard score copy. Exact replay is a surface-form reference; main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level FlatIP rebuild Soft O2 substantially improves Answer Hit over dense FlatIP; hard hydration is a score-copy envelope that can further raise AH by forcing siblings into top- k . Soft O2-C is evaluated as a conditional cluster-binding mechanism under Mix reconstructions with off-session gold; gated Hybrid is the composed operator when thematic co-membership helps. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

1. Introduction

2 A junior associate may ask a firm assistant: “Last month you advised
3 this client on terminating a fixed-term contract—what remedy did we rec-
4 ommend?” The system returns several near-identical “contract termination”
5 questions, ranks the stored user utterance highly, and never surfaces the part-
6 ner’s written advice. The subsequent reply may sound fluent yet be grounded

7 in the wrong episode—or in none. In consultation archives, questions and
8 answers share specialized terminology, neighbouring matters are topically ad-
9 jacent yet legally distinct, and the professionally useful evidence unit is the
10 full consultation *episode*.

11 Legal informatics has long studied statutory interpretation, case retrieval,
12 and advisory systems Martinez-Gil (2023); Ashley (2017), and recent legal
13 large language models (LLMs) further improve statutory and fact-pattern
14 answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical
15 legal information retrieval (IR), however, mainly targets *external* authority—
16 statutes, cases, or community answers—as documents Askari et al. (2024);
17 Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented
18 dialogue systems recover evidence from evolving conversation logs Wu et al.
19 (2024); Maharana et al. (2024); Packer et al. (2023), yet firm assistants still
20 need a reliable index of *internal consultation memory*: prior client-specific
21 advice buried in a growing same-domain archive under lexical interference
22 from neighbouring matters.

23 Under that interference, dense turn retrieval often returns a question
24 turn while its answer sibling remains outside top- k . We term this failure
25 *episode incompleteness*, as distinct from a full session miss. It is worsened
26 when approximate indexes distort turn scores and fragment episode context
27 at moderate archive sizes. Parent-document hydration and session-max ag-
28 gregation can force siblings into the result list by hard score copy, but they
29 also erase rank competition among unrelated candidates. What is needed is
30 a way to complete episodes without collapsing the retrieval ranking into a
31 session dump.

32 This paper presents BIMS-LEGAL, a dual-store system for Chinese legal
33 consultation logs. A fast episodic pathway indexes timestamped turn embed-
34 dings with speaker role and session identifier *sid*; a slower semantic pathway
35 consolidates turns with target- k KMeans (the production clustering path in
36 the released implementation). The dual-pathway split is inspired by Comple-
37 mentary Learning Systems theory McClelland et al. (1995); Kumaran et al.
38 (2016) only as an architectural analogy. On the episodic pathway, Soft O2
39 performs session soft binding: when a turn is retrieved with score s , sib-
40 lings sharing *sid* inherit $\max(s_t, \beta \cdot s)$, so an answer turn can surface under
41 question-hit / answer-miss while attenuated scores keep candidate competi-
42 tion intact. On the semantic pathway, Soft O2-C applies the same rule within
43 KMeans clusters when gold evidence is not co-session with the query; gated
44 Hybrid is intended to trigger cluster binding only on direct dense hits. Exact

FlatIP indexing (O1) restores faithful turn scores when product quantization collapses Answer Hit; bulk-load consolidation (O3) makes large shared-store construction tractable without changing retrieval semantics. We evaluate Soft O2 primarily against FlatIP under episode incompleteness, treat hard hydration as a score-copy envelope, and examine Soft O2-C under Mix re-constructions as a conditional mechanism test.

The main contributions are as follows. First, we formulate consultation-memory retrieval under same-domain interference and identify episode incompleteness as a recurrent failure mode of dense turn retrieval. Second, we propose Soft O2 as a turn-level soft-binding operator, with a gap-ratio account of attenuation β and default $\beta=0.98$ chosen by validation sweep, supported by O1/O3 as implementation operators for shared-store evaluation. Third, we evaluate Soft O2 on CAIL multi-turn and rebuilt LegalEp corpora against FlatIP, hard hydration, shuffled-*sid*, BM25-turn, RRF, and cross-encoder controls under a unified FlatIP-rebuild metric pipeline, and report Soft O2-C Mix contrasts as conditional mechanism evidence with graded nDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL and the Soft O2 attenuation analysis. Section 4 details corpora, metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent’s own archive. Confusing two clients’ histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM improve statutory and advisory generation Yue et al. (2023); Huang et al.

(2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives add a further difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question-answer episode. This setting motivates *dual-store* memory architectures that keep fast episodic turn retrieval separate from slower semantic clustering, and session-aware soft binding operators that recover answer turns under same-domain interference without collapsing rank competition among unrelated candidates.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL positions its dual-store contribution at this intersection: episodic turn embeddings with *sid* for same-session recovery (Soft O2), and KMeans semantic centroids with Soft O2-C for cross-session recovery when query and gold evidence occupy different sessions but share thematic structure. Under dense same-domain interference, a question turn can occupy top- k while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a

116 dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps
 117 ranking competition among candidates. Parent-document and hierarchical
 118 retrieval similarly attach child evidence to a parent unit Callan et al. (1994);
 119 Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2
 120 differs by using attenuated turn-level score propagation rather than hard hy-
 121 dration or atomic session documents. Lexical BM25 Robertson and Zaragoza
 122 (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Recipro-
 123 cal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira
 124 et al. (2019); Chen et al. (2024) provide complementary controls outside ses-
 125 sion membership. Exact flat indexes and product-quantized ANN backends
 126 Johnson et al. (2019) further affect whether sibling binding is applied over
 127 faithful turn scores.

128 *Contrast with joint-episode indexing..* An alternative way to keep episode
 129 context is to concatenate full question-answer pairs into monolithic episode
 130 documents (joint indexing). Joint indexing can prevent episode incomplete-
 131 ness by construction, but it changes the evidence unit from individual turns to
 132 merged documents. That coarser unit weakens fine-grained turn matching—
 133 for example, isolating a specific advice sentence—and raises index memory
 134 and update cost when new turns arrive in an online consultation log. Soft O2
 135 instead keeps turn-level representations and candidate competition, recover-
 136 ing missing siblings on demand via attenuated score inheritance (βs) with-
 137 out replacing the underlying turn index. Table 1 summarizes the session-
 138 structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

139 *On the reported corpora, session-max rankings coincide with hard parent hydration
 140 under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2
 141 denotes session soft binding; Soft O2-C denotes cluster soft binding on the KMeans se-
 142 mantic pathway.

143 In short, the dual-store split is the system design, and Soft O2 /Soft O2-C
 144 are the operators evaluated on Chinese legal consultation archives.

145 3. Method: BIMS-LEGAL

146 3.1. Architecture overview

147 Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complemen-
 148 tary episodic and semantic stores over the same consultation archive.

149 **Definition 1** (Episodic and semantic elements). *An episodic turn (episodic
 150 element) is $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a times-
 151 tamp, c_i conversational context (including speaker role), ϕ_i surface features,
 152 and sid_i the consultation session identifier. We reserve session turns for the
 153 weaker notion of turns that share only sid under a grouping key, without re-
 154 quiring role, timestamp, or other episodic metadata to participate in scoring.
 155 Soft O2 binds over session siblings $\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$, but
 156 the indexed objects themselves remain episodic turns. A semantic element is
 157 $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ with centroid $\mathbf{c}_j$, member set C_j , label κ_j , and formation
 158 statistics τ_j . Write $\text{Clus}(t) = \{t' : \text{cid}_{t'} = \text{cid}_t\} \setminus \{t\}$ for semantic-cluster
 159 siblings.*

160 **Definition 2** (Dual-store retrieval map). *Given query q with embedding
 161 $\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$
 162 under Eq. (1). Soft O2 produces $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$ (Sec-
 163 tion 3.3). Soft O2-C produces $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$ (Sec-
 164 tion 3.5). Gated Hybrid composes session binding then cross-session cluster
 165 binding (Algorithm 2). The returned list is the top- k of the bound set under
 166 the rewritten scores.*

167 3.2. Base scoring and implementation operators (O1, O3)

168 For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory
 169 turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

170 where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

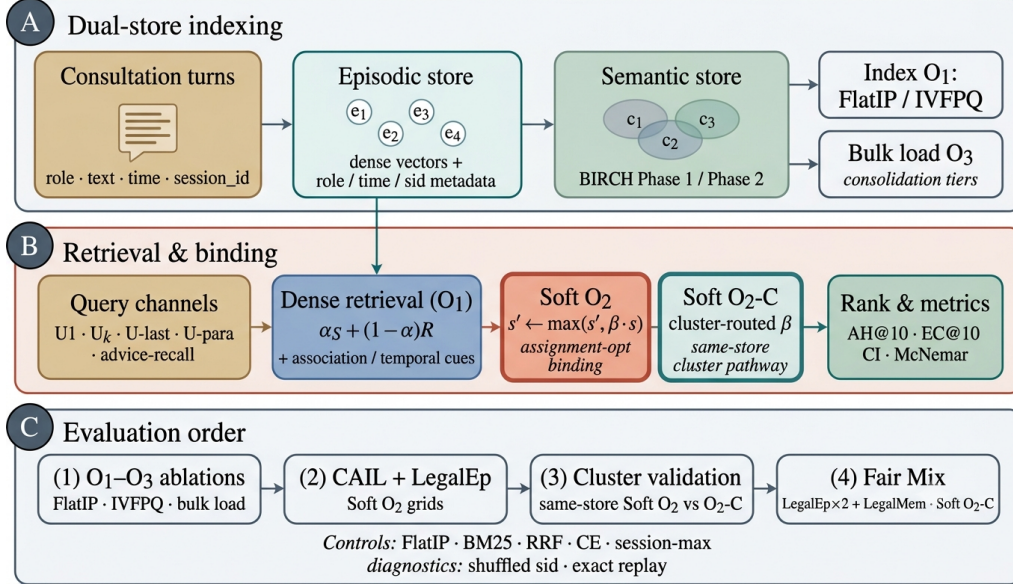


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and KMeans semantic consolidation (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

171 with Δ_t the elapsed time between query time and the turn timestamp. Unless
 172 stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the
 173 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step
 174 and do not replace Eq. (1).

175 *O1: Exact FlatIP indexing..* Let $\hat{S}$ denote the similarity induced by an ANN
 176 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-
 177 trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2
 178 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 179 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 180 ble A.3: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 181 engineering prerequisite for Soft O2 evaluation, not a standalone retrieval
 182 algorithm.

183 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 184 insert update ($e_i \mapsto \text{KMeansConsolidate}(e_i); \text{Flush}$) by a deferred operator
 185 $\text{Finalize} = \text{KMeansConsolidate}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic

186 writes. O3 does not change retrieval semantics; it makes $M \geq 400$ shared-
 187 store experiments runnable (≈ 11 min wall-clock for $M=400$; amortized $\approx$
 188 0.825 s/turn over ~ 800 turns). Quality–latency curves to $M=6400$ appear
 189 in Section 5.6.

190 3.3. Soft O2: session soft binding

191 *Motivation..* Under same-domain interference, a retrieved question turn can
 192 score highly while its answer sibling remains outside top- k —episode incom-
 193 pleteness compounded by IVFPQ score collapse and context fragmentation.
 194 Soft O2 completes episodes without hard score copy.

195 **Definition 3** (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a*
 196 *sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

197 *Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is*
 198 *the special case $\beta=1$ with forced insertion of all siblings; session-max replaces*
 199 *each member score by $\max_{t \in \text{sid}} s_t$.*

200 The released Soft O2 pipeline expands session siblings on raw dense scores,
 201 optionally applies exact-match boost and sparse augment, performs session-
 202 first reranking, fuses semantic and recency scores (Eq. (1)), then runs a final
 203 completeness pass before top- k truncation. Algorithm 1 abstracts the binding
 204 core (expansion + completeness); β therefore acts on the score field used at
 205 each expansion stage as implemented in `memory_manager.py`.

206 3.4. Soft O2 attenuation and score competition

207 The attenuation β controls how strongly a retrieved turn promotes its
 208 session siblings. Availability of the injected sibling rises with β , while fidelity
 209 of the direct hit falls as $\beta \rightarrow 1$. Table 2 summarizes the empirical gap-ratio
 210 distribution $\rho = s_d/s_h$ on LegalEp stores under an exact-channel query proxy:
 211 median ρ lies well below one, and a high- β operating point near 0.98 covers
 212 most measured distractors while remaining strictly below hard hydration
 213 ($\beta=1$). We therefore characterize the availability–ranking trade-off with a
 214 three-candidate score-competition model and choose β by validation sweep
 215 rather than by a closed-form optimum. A zero-margin hinge objective on
 216 $\beta \in (0, 1]$ is uninformative for this choice: its rank term vanishes for all
 217 $\beta < 1$.

Algorithm 1 Soft O2 session soft binding

Require: Query q , base hits $\mathcal{R}_0$, top- k , attenuation β

```
1:  $\mathcal{R} \leftarrow \mathcal{R}_0$ 
2: for each hit  $t \in \mathcal{R}_0$  with score  $s_t$  do
3:   for each sibling  $t' \in \text{Sib}(t)$  absent from  $\mathcal{R}$  do
4:     insert  $(t', \beta \cdot s_t)$  into  $\mathcal{R}$ 
5:   end for
6: end for
7:                                      $\triangleright$  Completeness pass after fusion with Eq. (1)
8: for each session  $sid$  represented in  $\mathcal{R}$  do
9:    $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R}, sid_u = sid\}$ 
10:  for each  $t' \in \text{Sib}$  of  $sid$  still absent do
11:    insert  $(t', \beta \cdot s_{\max})$  into  $\mathcal{R}$ 
12:  end for
13: end for
14: return top- $k$  of  $\mathcal{R}$  by descending score
```

Table 2: Gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). Default β is chosen by the AH sweep in Table A.4.

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

218 **Definition 4** (Three-candidate score-competition model). *For a query whose*
219 *dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be*
220 *the gold answer sibling and let d be the strongest non-session distractor in a*
221 *wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

222 *and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured*
223 *distractor, conditions based on ρ are conservative sufficient conditions for*
224 *beating that distractor, not exact top- k boundary conditions for every com-*
225 *petitor.*

226 The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$,
227 where $s'_a = \beta s_h$.

228 **Proposition 1** (Feasible attenuation band). *Under Definition 4, assume*
 229 *the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured*
 230 *distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the*
 231 *injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$,*
 232 *any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$*
 233 *already, rank preservation depends on the observed sibling score rather than*
 234 *on β alone.*

235 *Proof.* If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e.
 236 $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when
 237 $s_h > 0$. $\square$

238 *Soft-rank visualization and sweep.* On a validation grid we also plot the soft
 239 pairwise surrogate

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

240 with logistic σ , temperature τ , and rank weight γ . The curve is a visualization
 241 of the availability–ranking trade-off under the measured ρ pool; it is not used
 242 as a fitted optimum. Default $\beta=0.98$ is taken from the AH sweep in Table A.4,
 243 where $\{0.90, 0.95, 0.98\}$ form a stable high-AH band on CAIL follow-up and
 244 LegalEp paraphrase as well as exact grids. Advice-recall uses the same default
 245 without a separate gap-ratio fit.

246 Fair controls include hard parent hydration, session-max aggregation, and
 247 shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max
 248 coincide under full-sibling expansion, so tables report a single hard-expansion
 249 baseline. Concatenating a full question–answer episode into one retrieval unit
 250 (BM25-joint or dense joint-episode documents) changes the evidence unit
 251 from turns to documents; we treat that design as an alternative indexing
 252 choice rather than a Soft O2 turn-binding baseline (Section 4).

253 3.5. Semantic consolidation and Soft O2-C

254 Semantic consolidation in the released system uses target- k KMeans over
 255 turn embeddings (`encode_with_target_k`), producing cluster ids consumed
 256 by Soft O2-C. An optional sklearn Birch fit remains in the codebase only as
 257 an unused ablation helper and is not on the Soft O2-C evaluation path.

258 **Definition 5** (KMeans semantic consolidation). *Given embeddings $\{e_i\}$ and*
 259 *target cluster count k , O3 runs $(\text{labels}, \text{centers}) \leftarrow \text{KMeans}(\{e_i\}; k)$ once at*

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

- 1: $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$ $\triangleright$ session pathway first
- 2: $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$
- 3: $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$ $\triangleright$ direct dense hits only
- 4: **for** each trigger $t \in \mathcal{T}_{\text{trig}}$ **do**
- 5: $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$
- 6: **for** each $t' \in \text{Clus}(t)$ with $sid_{t'} \notin \mathcal{S}_{\text{cov}}$, up to budget B_c **do**
- 7: $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$
- 8: **end for**
- 9: **end for**
- 10: **return** top- k of $\mathcal{R}$

260 *finalize time and assigns each turn t a cluster id cid_t , inducing $\text{Clus}(t)$ in Def-*
261 *inition 1. Empty clusters are dropped and labels are remapped to a contiguous*
262 *range.*

263 **Definition 6** (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default*
264 *$\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session mem-*
265 *bership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large*
266 *thematic clusters cannot flood top- k .*

267 Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings
268 relative to sessions already covered by dense /Soft O2 hits, and triggers
269 cluster binding only on direct dense hits. This prevents thematic confusers
270 from displacing same-session siblings already recovered by Soft O2. Soft O2
271 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway
272 operator when gold evidence lies in another session of the same thematic
273 cluster.

274 4. Corpora and Evaluation Protocol

275 Figure 2 sketches the shared-corpus design used throughout the legal eval-
276 uation. Gold needles and same-domain distractors occupy one store. Each
277 query probes that shared archive under fixed interference conditions. Pri-
278 vately rebuilt haystacks are avoided so that system comparisons isolate the
279 retrieval operator.

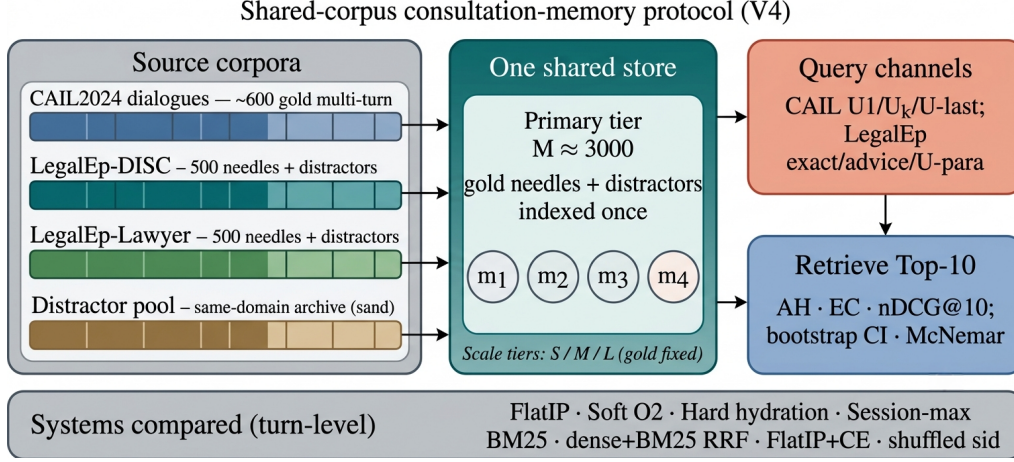


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

280 4.1. Datasets and query channels

281 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
 282 Gold needles are authentic preliminary and final dialogues, with conversation
 283 turns completed by label turns where applicable. Query channels cover the
 284 first user turn ($U1$), a later in-dialogue user turn ($U_k\text{-followup}$), the last user
 285 turn ($U\text{-last}$), and constrained paraphrases ($U\text{-para}$). Evidence defaults to
 286 assistant turns in the target session.

287 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
 288 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
 289 a natural (question, answer) pair with session identifier *sid*. Reconstruction
 290 applies length bounds and legal-content filters, excludes exam prompts, re-
 291 moves near-duplicates, separates needles from distractors under a fixed seed,
 292 and retains an audit subsample. After filtering, each LegalEp corpus retains
 293 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary
 294 tier).

295 *Query-channel policy.* Exact replay uses the stored user question and is
 296 reported only as a *diagnostic* upper bound on surface-form recovery. Primary
 297 claims use paraphrase, $U_k\text{-followup}$ / $U\text{-last}$, and advice-recall queries whose
 298 surface form is not identical to the stored question text; paraphrase and
 299 advice-recall templates are generated under a fixed seed and audited on a

subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions. Archive size is varied with gold needles held fixed. Unless stated otherwise, Soft O2 transfer tables use the $M \approx 3000$ tier; O1-O2 ablations use $M=400$, $S=300$.

4.2. Metrics and baselines

Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC), and normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is binary: whether a gold answer turn appears in the top- k . EC is the mean coverage of gold turns within the target session. Session Hit@ k is reported only for diagnosis. For nDCG we use fixed graded relevance over the full gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all gold-relevant turns in the target session and truncated at k , so every system on a query shares the same IDCG. Answer Hit and Episode Completeness are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session IDCG so that systems remain comparable on each query. Hard expansion can raise nDCG when copied siblings occupy early ranks, so AH and EC are read together with nDCG.

The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL and LegalEp transfer channels; for that family we report McNemar mid- p on paired per-query Answer Hit and Holm-adjusted p -values (Table A.2). Appendix CAIL grids report AH point estimates from the same rebuild; LegalEp appendix grids report AH/EC/nDCG without separate CI columns. Hard hydration, CE, and Soft O2-C Mix contrasts are secondary.

Unless noted, Soft O2 grids in Sections 5.3–5.4 and Appendix Tables A.5–A.7 use one FlatIP rebuild so that AH, EC, nDCG, and failure rates on each row come from the same per-query rankings (`corrected_metrics_*.json` under `paper/ipm/figures/`).

Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies soft sibling score inheritance on the episodic pathway; Soft O2-C applies the same rule within KMeans clusters. Hard parent hydration expands a hit to all siblings by copying the trigger score. Lexical controls include BM25 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve pre-concatenated episode documents and therefore answer a coarser retrieval problem; they are discussed as alternative designs in Section 6.3 rather than

337 Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the
 338 top-30 FlatIP candidates with BAAI/bge-reranker-v2-m3. Shuffled *sid*,
 339 a β sweep, and IVFPQ provide membership, attenuation, and quantization
 340 controls.

341 4.3. Fair Mix protocol for Soft O2-C

342 On fully same-session LegalMem grids Soft O2 already recovers gold an-
 343 swers that share *sid*, leaving Soft O2-C little exclusive headroom and expos-
 344 ing it to thematic confusers. To study off-session recovery we rebuild the *same*
 345 LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes
 346 are split into query-side S_a and evidence-side S_b with distinct session identi-
 347 fiers, while 30% remain intact. Distractors are unchanged. FlatIP, Soft O2,
 348 Soft O2-C, and gated Hybrid share unrestricted dense retrieval on this store;
 349 we report overall AH and stratum AH on `cross_session` vs `same_session`
 350 queries.

351 *Cluster co-membership and purity under Mix..* After KMeans consolidation,
 352 each cross-session Sa/Sb pair is assigned a shared cluster identifier. This
 353 construction uses split metadata available only when building the evaluation
 354 store; it does not inject labels at query time. The intent is to test Soft O2-
 355 C when thematic co-membership is present. Cluster purity still matters: a
 356 shared *cid* that also contains many thematically adjacent distractors narrows
 357 Soft O2-C’s effective margin, which is why Hybrid gates cluster injection to
 358 direct dense hits and caps the per-cluster budget B_c . If recovery still fails
 359 under guaranteed Sa/Sb co-membership, the failure is attributable to binding
 360 rather than accidental cluster separation. Natural same-cluster rates without
 361 this construction are lower (Section 6.3). Soft O2’s session membership rule
 362 is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses
 363 mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior
 364 runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon
 365 memory context only Wu et al. (2024); Maharana et al. (2024).

366 5. Experiments

367 Unless noted, all dense retrieval grids use the multilingual encoder `nomic-embed-text-v2-moe`
 368 Nussbaum et al. (2025) (768-d) as a single shared embedding backend, with
 369 $k=10$ and seed 42 on the main $M \approx 3000$ stores. We keep the encoder fixed
 370 so Soft O2 vs FlatIP contrasts isolate binding rather than representation

changes; sensitivity to alternative Chinese or multilingual encoders (e.g., `bge-large-zh-v1.5`) is discussed in Section 6.3. A five-seed check on a smaller shared store ($M=400$) is reported later. Primary contrasts appear in Figures 3–5; full numeric grids are in the appendix. Each main table maps to a released JSON artifact in the accompanying repository.

5.1. Implementation operators and Soft O2 ablation

Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirming that exact indexing is a necessary implementation prerequisite but insufficient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283 over baseline) and Episode Completeness to 0.888 / 0.840. O3 is enabled for all configurations in this ladder as a construction prerequisite; it does not alter retrieval scores.

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M \approx 3000$ grids appears in Table 6.

Corpus	System	EC@10	AH@10
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	0.888	0.780
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	0.840	0.680

5.2. Failure taxonomy

We classify each query into four mutually exclusive modes: *complete* (question and answer in top- k), *incomplete* (question hit / answer miss), *answer-only* (answer hit / question miss), and *session miss* (no gold-session turn in top- k). Table 4 reports these rates for FlatIP, Soft O2, hard hydration, and shuffled *sid* on primary non-exact channels under the unified FlatIP rebuild. Soft O2 converts many incomplete FlatIP cases into complete retrieval while session-miss stays nearly fixed; shuffled *sid* restores high incompleteness. Hard hydration can drive incompleteness near zero by score copy and

Table 4: Failure taxonomy ($k=10$) with four mutually exclusive classes from per-query labels: complete (Q+A in top- k), incomplete (Q hit / A miss), answer-only (A hit / Q miss), and session miss. Boldface: best complete / lowest incomplete within each block among FlatIP, Soft O2, and shuffled *sid* (hard hydration shown as a score-copy envelope). Rates sum to 100% within rounding.

Corpus / channel	System	Complete	Incomplete	Ans-only	Session miss
LegalEp-DISC / u-para	FlatIP	70.8%	26.8%	0.8%	1.6%
LegalEp-DISC / u-para	Soft O2	89.1%	8.7%	0.4%	1.8%
LegalEp-DISC / u-para	Hard hydr.	97.0%	0.0%	0.2%	2.8%
LegalEp-DISC / u-para	Shuffled O2	66.0%	31.4%	0.4%	2.2%
LegalEp-DISC / advice	FlatIP	48.0%	17.0%	0.0%	35.0%
LegalEp-DISC / advice	Soft O2	60.8%	3.8%	0.0%	35.4%
LegalEp-DISC / advice	Hard hydr.	64.8%	0.0%	0.2%	35.0%
LegalEp-DISC / advice	Shuffled O2	45.6%	19.2%	0.0%	35.2%
LegalEp-Lawyer / u-para	FlatIP	71.5%	25.3%	2.4%	0.8%
LegalEp-Lawyer / u-para	Soft O2	94.0%	3.8%	0.6%	1.6%
LegalEp-Lawyer / u-para	Hard hydr.	98.0%	0.0%	0.0%	2.0%
LegalEp-Lawyer / u-para	Shuffled O2	70.7%	25.5%	2.2%	1.6%
CAIL / Uk-followup	FlatIP	45.8%	54.0%	0.0%	0.2%
CAIL / Uk-followup	Soft O2	92.8%	7.0%	0.0%	0.2%
CAIL / Uk-followup	Hard hydr.	98.3%	1.5%	0.0%	0.2%
CAIL / Uk-followup	Shuffled O2	34.7%	65.2%	0.0%	0.2%

often leads Soft O2 on complete rate; Soft O2 remains the attenuated binder ($\beta < 1$) rather than a hard session dump (Section 5.4).

5.3. Turn-level Soft O2 controls

Primary Soft O2 claims use a shared turn-level index. Table 5 reports Answer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the primary Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact channel; shuffled *sid* collapses toward or below FlatIP. Hard hydration—full sibling score copy—can exceed Soft O2 AH on the same rebuild (Table 5); Soft O2 is therefore best read as an attenuated binder that recovers missing answers without the unrestricted score-copy envelope of hard expansion (Tables A.5–A.7). BM25-turn is a same-granularity lexical control (Table A.8). Joint episode / document-level indexes change evidence granularity and are scoped as alternative designs rather than Soft O2 turn-binding baselines (Section 6.3).

Table 5: Primary Soft O2 controls (AH@10, $M \approx 3000$, unified FlatIP rebuild). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is compared primarily to FlatIP; hard hydration is the score-copy envelope; shuffled *sid* tests membership dependence.

Corpus / channel	FlatIP	Soft O2	Hard	Shuffled
CAIL / U1	0.832	0.960	0.998	0.695
CAIL / Uk	0.458	0.928	0.983	0.347
CAIL / U-last	0.425	0.945	0.983	0.287
LegalEp-DISC / u-para	0.716	0.895	0.972	0.664
LegalEp-DISC / advice	0.480	0.608	0.650	0.456
LegalEp-Lawyer / u-para	0.739	0.946	0.980	0.729
LegalEp-Lawyer / advice	0.444	0.652	0.674	0.424

5.4. Soft O2 on session (CAIL and LegalEp)

Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP on every channel under the unified rebuild (U1 0.960 vs 0.832; Uk 0.928 vs 0.458; U-last 0.945 vs 0.425). Shuffled *sid* removes the gain, tying the improvement to correct episode membership. Hard expansion can raise AH further by unrestricted score copy (e.g., Uk 0.983 vs Soft O2 0.928). Soft O2 is therefore the attenuated binder: siblings enter under $\beta < 1$ scores, whereas hard hydration is the score-copy envelope. Table 6 shows the Soft O2 > FlatIP ordering under graded nDCG with gold-session IDCG, while incompleteness falls sharply (Table 4). Full numeric grids appear in Tables A.5–A.7.

Figure 4 reports LegalEp transfer on the same AH grids. On LegalEp-DISC exact replay Soft O2 and FlatIP stay near-tied (0.638 vs 0.640), where both turns are already recoverable for many queries—consistent with treating exact as diagnostic. On Lawyer exact Soft O2 improves AH to 0.752 from FlatIP 0.692 (McNemar mid- $p=0.024^*$). Paraphrase yields large Soft O2 gains over FlatIP: 0.895 vs 0.716 on DISC and 0.946 vs 0.739 on Lawyer under the unified rebuild. Advice-recall similarly rises to 0.608 from FlatIP 0.480 on DISC and to 0.652 from 0.444 on Lawyer. Hard hydration can still edge Soft O2 by unrestricted score copy on these channels; Soft O2 remains the attenuated alternative ($\beta < 1$). Shuffled *sid* collapses toward FlatIP. Graded nDCG mirrors the Soft O2 > FlatIP ordering on paraphrase and advice-recall (Table 6). Detailed grids are in Tables A.6–A.7.

Cross-encoder reranking is a strong neural control at the same turn-

Table 6: Graded nDCG@10 with gold-session IDCG (answer $rel=1.0$, other same-session turns $rel=0.5$) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.445	54.0%
CAIL	Uk	Soft O2	0.793	7.0%
CAIL	Uk	Shuffled O2	0.369	65.2%
CAIL	U-last	FlatIP	0.420	57.3%
CAIL	U-last	Soft O2	0.791	5.3%
CAIL	U-last	Shuffled O2	0.348	71.2%
LegalEp-DISC	u-para	FlatIP	0.681	26.8%
LegalEp-DISC	u-para	Soft O2	0.759	8.7%
LegalEp-DISC	u-para	Shuffled O2	0.627	31.4%
LegalEp-DISC	advice	FlatIP	0.466	17.0%
LegalEp-DISC	advice	Soft O2	0.524	3.8%
LegalEp-DISC	advice	Shuffled O2	0.436	19.2%
LegalEp-Lawyer	u-para	FlatIP	0.744	25.3%
LegalEp-Lawyer	u-para	Soft O2	0.844	3.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.705	25.5%
LegalEp-Lawyer	advice	FlatIP	0.483	23.4%
LegalEp-Lawyer	advice	Soft O2	0.582	2.2%
LegalEp-Lawyer	advice	Shuffled O2	0.454	25.0%

432 level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with
 433 **bge-reranker-v2-m3**. Table 7 shows that FlatIP+CE is competitive—and
 434 on LegalEp-DISC exact, strongest—when surface-form overlap already places
 435 both question and answer near the top of the dense pool (DISC exact AH
 436 0.722 for FlatIP+CE vs 0.638 for Soft O2). On channels with semantic shift
 437 or multi-turn drift, Soft O2 pulls ahead: CAIL Uk 0.698 vs FlatIP+CE 0.375,
 438 CAIL U-last 0.762 vs 0.342, and LegalEp-Lawyer exact 0.752 vs 0.708. A
 439 structural limit of the reranker explains the split: CE can refine ranking
 440 among candidates already retrieved by FlatIP, but it cannot promote an
 441 answer sibling that never entered the top-30 pool. Soft O2 recovers such
 442 siblings by propagating attenuated scores through session links, which helps
 443 precisely when the stored advice turn has weak direct lexical overlap with
 444 the query (paraphrase and advice-recall; cf. Tables 5 and 6).

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

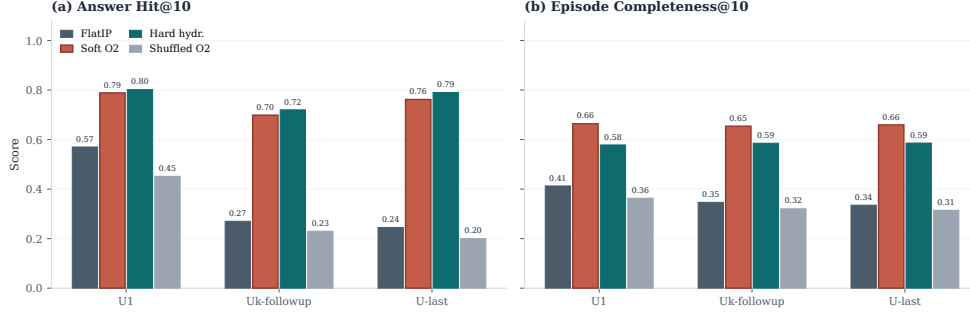


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

Table 7: Cross-encoder control (AH@10). FlatIP+CE reranks the top-30 FlatIP candidates with **bge-reranker-v2-m3**. Boldface: best AH within each row.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.638
LegalEp-Lawyer / exact	0.636	0.708	0.752

5.5. Soft O2-C and gated Hybrid under Mix

When gold answers already share *sid* with the query, Soft O2 is sufficient and cluster binding is unnecessary. Table 8 asks the off-session question under the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated Hybrid sharing unrestricted dense retrieval. Hybrid uses the post-gate direct-dense trigger: cluster binding fires only on tids recorded *before* Soft O2 session expansion.

On LegalEp Mix stores, gated Hybrid is the strongest overall system (DISC AH 0.542, Lawyer 0.536; McNemar mid- $p=0.0005^{**}$ / 0.0022^{**} vs Soft O2) and also leads AH_{cross} . Ungated Soft O2-C raises DISC AH_{cross} to 0.506 from Soft O2 0.460 and Lawyer cross to 0.480 from 0.440, but remains weaker than Hybrid once session binding is composed under the gate. On LegalMem-MT Mix the picture reverses: Soft O2 retains the best overall AH (0.602), while Soft O2-C (0.440) and Hybrid (0.562) trail; FlatIP

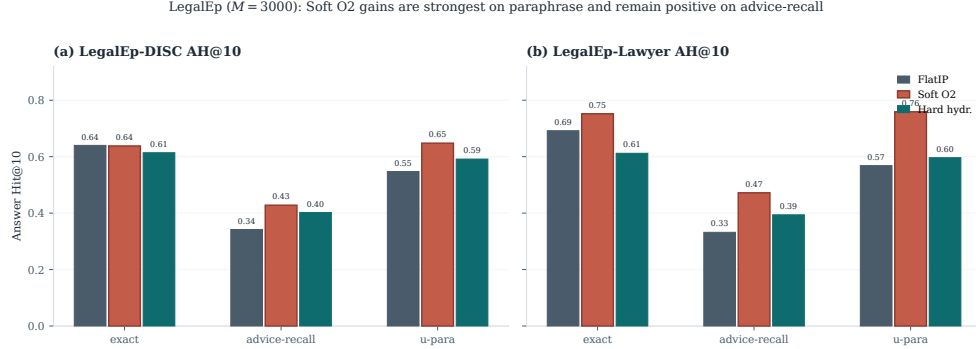


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

even leads the cross stratum (0.520). Cluster binding is therefore conditional on corpus and cluster purity—helpful as a gated add-on on LegalEp advice-recall Mix, but not a free lunch on CAIL multi-turn Mix under the same protocol. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led (Appendix Appendix A, Table A.1).

Hybrid uses the post-gate direct-dense trigger. Soft O2-C mid- p vs Soft O2 may favour either side; on LegalMem-MT the ** marks Soft O2-C significantly *below* Soft O2. Cluster co-membership of Sa/Sb pairs follows Section 4.3.

5.6. Scale behaviour

Figure 5 and Table A.3 report quality and warm-query latency on LegalEp-DISC exact ($Q=100$, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2 retains an AH advantage at each tier; IVFPQ collapses at $M=1600$ (AH 0.330), motivating O1; absolute latency grows with M . We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and avoid claiming deployment readiness beyond the measured curve.

End-to-end QA audit (summary).. A dual-protocol QA audit ($N=270$ per corpus; generator **qwen3:14b**, independent judge **qwen3:32b**) reports judged answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs Soft O2 generation experiment. Full protocol and stratified results appear in Section 6.2. On a smaller shared store ($M=400$, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

Table 8: Mix evaluation of Soft O2-C and gated Hybrid after the direct-dense gate fix (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p<0.05$, ** $p<0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.494	0.491	0.500	—
	Soft O2	0.492	0.460	0.567	—
	Soft O2-C	0.498	0.506	0.480	0.7275
	Hybrid	0.542	0.526	0.580	0.0005**
LegalEp-Lawyer Mix	FlatIP	0.460	0.451	0.480	—
	Soft O2	0.492	0.440	0.613	—
	Soft O2-C	0.464	0.480	0.427	0.1284
	Hybrid	0.536	0.520	0.573	0.0022**
LegalMem-MT Mix	FlatIP	0.530	0.520	0.553	—
	Soft O2	0.602	0.489	0.867	—
	Soft O2-C	0.440	0.460	0.393	0.0000**
	Hybrid	0.562	0.446	0.833	0.0594

6. Discussion

6.1. Findings and scope

Results show that episode incompleteness is a substantial FlatIP failure mode under same-domain interference, and that Soft O2 converts many incomplete retrievals into complete ones without shrinking session-miss rates (Table 4). O1 restores score fidelity; Soft O2 supplies the main gain through soft sibling inheritance; O3 makes large shared stores tractable. The gap-ratio analysis (Section 3.4) motivates keeping β close to but below unity; the AH sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration ($\beta=1$). On CAIL under the unified rebuild, Soft O2 raises Answer Hit by large margins over FlatIP (Uk 0.928 vs 0.458; U-last 0.945 vs 0.425) with collapsing shuffled-*sid* controls. Hard hydration can exceed Soft O2 AH/EC as a score-copy envelope. After Holm adjustment on the Soft O2 vs FlatIP family, the non-exact CAIL and LegalEp gains remain significant, while LegalEp exact replay is mixed (Table A.2). LegalEp paraphrase and advice-recall, rather than exact replay, support the main transfer result. Graded nDCG agrees with the Soft O2>FlatIP ordering (Table 6). Under Mix, gated Hybrid significantly improves overall AH over Soft O2 on LegalEp-DISC and LegalEp-Lawyer, while Soft O2 remains strongest on LegalMem-MT (Table 8); Soft O2

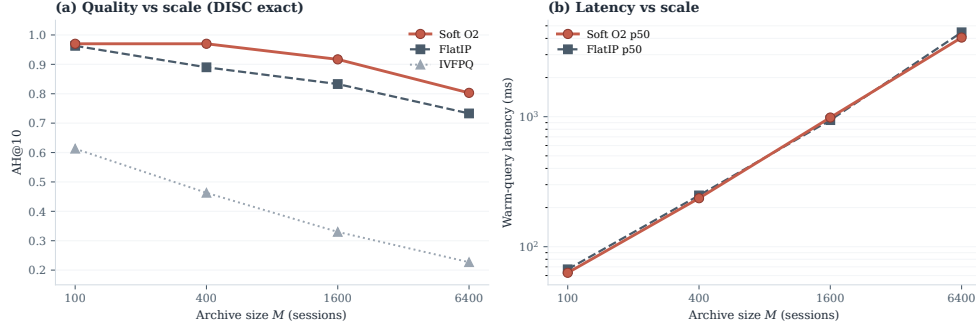


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

599 stays preferable when gold already shares *sid* (Table A.1). Turn-level BM25,
 500 RRF, and CE place Soft O2 among standard IR alternatives Robertson and
 501 Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode
 502 indexes are a coarser design choice outside this protocol.

503 6.2. RAG application perspective

504 Consultation-memory retrieval is judged downstream by whether an LLM
 505 can answer from recovered evidence rather than confabulate. We audit this
 506 end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: genera-
 507 tor `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per
 508 corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$ (folder
 509 `legal_qa_n270`). The independent judge assigns a continuous evidence-
 510 answerability score in $[0, 1]$ from the retrieved context and the generated
 511 reply; Evidence-Answerability Rate (EAR) is the fraction of instances with
 512 judge score ≥ 0.7 . Table 9 reports pooled EAR and the mean judge score
 513 on a stratified subsample ($n=90$ /corpus; 30 each of exact, paraphrase, and
 514 follow-up), labelled “Stratified judge” in the table. In the complete-episode
 515 stratum—both question and answer turns in top- k —judged answerability is
 516 higher in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on
 517 Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain
 518 the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with
 519 episodic incompleteness persisting even when surface-form overlap is low.

520 A blind two-annotator legal audit ($n=60$ /corpus) separates retrieval fail-
 521 ures (gold episode absent from top- k) from generation failures given retrieved
 522 evidence. Annotators used a binary answerability label on each instance;

agreement is Cohen’s κ computed on those paired labels (0.71 on DISC, 0.68 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 9: End-to-end QA audit ($N=270$ per corpus; generator **qwen3:14b**, independent judge **qwen3:32b**). Pooled EAR: fraction with judge score ≥ 0.7 . Stratified judge: mean judge score on $n=90$ /corpus (exact/paraphrase/follow-up). Human κ : Cohen’s κ on binary answerability labels from two blind annotators ($n=60$ /corpus).

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL.

Dependency on session segmentation and cluster quality.. Soft O2 relies on correct session identifiers (*sid*). In uncured client logs, boundary noise—mid-session topic shifts, multi-intent interruptions, or automated segmentation drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated inheritance may propagate false context; if a true advice turn is split into another session, Soft O2 cannot bind it. Soft O2-C likewise depends on KMeans cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared cluster id so binding can be tested when thematic co-membership is present (Section 4.3); in fully unsupervised consolidation, noisier boundaries shrink the margin over dense retrieval. Deployments on raw consultation streams therefore need reliable upstream session segmentation, and Soft O2-C should be treated as gated rather than always-on.

548 *Joint-episode indexing..* Concatenating full question–answer pairs into
 549 episode documents prevents incompleteness by changing the evidence unit
 550 from turns to merged documents (Section 2). That design is outside the
 551 Soft O2 turn-binding protocol; a matched comparison under online updates,
 552 latency, and fine-grained advice isolation is left to future work.

553 *Embedding model and other scope limits..* All main grids use
 554 `nomic-embed-text-v2-moe`. A Chinese-specialized encoder such as
 555 `bge-large-zh-v1.5` could alter absolute dense ranks and the measured ρ
 556 distribution, and might shrink or widen Soft O2’s relative gain wherever
 557 the bi-encoder already places answer turns inside top- k . We did not re-run
 558 the full Soft O2 grids under alternative encoders; the binding operator
 559 itself is encoder-agnostic once fused scores are available. The ρ statistics
 560 use an exact-channel query proxy and the strongest measured distractor;
 561 paraphrase and advice-recall gap laws may shift the feasible band slightly,
 562 though the AH sweep keeps a stable high- β region on those channels. Main
 563 $M \approx 3000$ tables use a fixed seed; five-seed variance is reported only for the
 564 smaller $M = 400$ store. LLM paraphrases need human audit. Statutory
 565 correctness, citation validity, and professional liability remain unevaluated.
 566 Because many channels are tested, Holm-adjusted p -values accompany the
 567 Soft O2 vs FlatIP family (Table A.2); Mix Hybrid significance is reported
 568 only vs Soft O2 within Table 8 and is corpus-conditional.

569 *Latency and deployment scope..* Scale curves cover $M \in$
 570 $\{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.3). At
 571 $M = 6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms
 572 for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL
 573 should therefore be positioned as **high-precision offline or quasi-**
 574 **real-time consultation memory recovery**—e.g., pre-meeting case
 575 review, audit replay, or batch RAG over a firm’s advice archive—not as
 576 a sub-millisecond search engine. Future work will combine approximate
 577 nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C
 578 binding layered on top of faithful candidate retrieval, preserving episode
 579 completion while reducing query latency. Behavior at roughly 10^5 sessions is
 580 not measured, and we do not claim asymptotic scalability from O3 wall-clock
 581 alone.

582 7. Conclusion

583 BIMS-LEGAL recovers prior advice from Chinese legal consultation
584 archives under same-domain interference with a dual store: an episodic turn
585 index and a KMeans semantic store. Soft O2 is the main algorithmic con-
586 tribution: attenuated session binding completes episodes without hard score
587 copy, while hard hydration remains a stronger score-copy envelope on the
588 unified rebuild grids. Soft O2-C extends the same idea to thematic clusters
589 under Mix reconstructions as a conditional mechanism test. Attenuation β
590 is chosen by validation sweep under a three-candidate score-competition ac-
591 count; O1 FlatIP and O3 bulk-load are the implementation operators that
592 make the shared-store ablations feasible. On CAIL multi-turn and LegalEp
593 paraphrase /advice-recall channels, Soft O2 improves Answer Hit over FlatIP,
594 keeps higher episode completeness than hard expansion on CAIL, and raises
595 graded nDCG while cutting incompleteness; shuffled-*sid* controls collapse the
596 gains. Under Mix, gated Hybrid recovers additional LegalEp gold when clus-
597 ter co-membership is present, while Soft O2 remains preferable on LegalMem-
598 MT Mix and on intact same-session gold. An end-to-end Soft O2 QA au-
599 dit associates complete episode retrieval with judged answerability and low
600 wrong-session contamination, without claiming paired generation superiority
601 over FlatIP. Turn-level BM25, RRF, and CE place Soft O2 among standard
602 IR alternatives; joint-episode indexing remains a coarser design outside this
603 protocol.

604 Declarations

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606 Computational resources were provided by the Data Law Laboratory,
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608 *Conflict of interest*

609 The authors declare no competing interests.

610 *Ethics approval*

611 This study uses publicly released research corpora and excludes confiden-
612 tial client data. The system is a research prototype for consultation-memory
613 retrieval.

614 *Data availability*

615 Code, processed evaluation corpora, and primary result JSON files
616 will be made available upon acceptance. Upstream corpora remain un-
617 der their original licenses: CAIL2024 consultation tracks; Hugging Face
618 ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_data. Processed
619 LegalEp /LegalMem artifacts are derived research releases; redistributors
620 should consult each upstream licence before reuse.

621 *Code availability*

622 The BIMS-LEGAL implementation, evaluation pipelines, and ta-
623 ble/figure scripts will be released with the accepted publication.

624 *Author contributions*

625 Author contributions are withheld for anonymous review.

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740 **Appendix A. Numeric grids and controls**

741 Same-session Soft O2-C contrasts, significance tests, scale curves, and full
 742 Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

Table A.2: Primary Soft O2 vs FlatIP family: McNemar mid-*p* on paired Answer Hit from V4 `per_query_ah`, with Holm step-down adjustment ($m=9$). Artifact: `paper/ipm/figures/holm_primary_family.json`.

Corpus / channel	Contrast	ΔAH	mid- <i>p</i>	Holm <i>p</i>
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	7.19e-47
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	3.63e-17
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.9341	0.9341
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.02386	0.04773
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	3.18e-04
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.26e-14
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	3.76e-04
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	5.89e-10

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503

Table A.4: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.832	0.554	0.554	—
	Soft O2	0.960	0.924	0.776	—
	Hard hydr.	0.998	0.997	0.864	—
	Shuffled O2	0.695	0.437	0.411	—
Uk-followup	FlatIP	0.458	0.451	0.445	—
	Soft O2	0.928	0.923	0.793	—
	Hard hydr.	0.983	0.979	0.847	—
	Shuffled O2	0.347	0.367	0.369	—
U-last	FlatIP	0.425	0.433	0.420	—
	Soft O2	0.945	0.919	0.791	—
	Hard hydr.	0.983	0.977	0.854	—
	Shuffled O2	0.287	0.359	0.348	—

743

Soft O2 leads EC; hard hydration can lead AH.

Table A.6: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p < 0.05$, ** $p < 0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.742	0.871	0.730
	Soft O2	0.924	0.962	0.813
	Hard hydr.	1.000	1.000	0.886
	Shuffled O2	0.682	0.841	0.666
advice-recall	FlatIP	0.480	0.565	0.466
	Soft O2	0.608	0.627	0.524
	Hard hydr.	0.650	0.649	0.572
	Shuffled O2	0.456	0.552	0.436
u-param	FlatIP	0.716	0.846	0.681
	Soft O2	0.895	0.937	0.759
	Hard hydr.	0.972	0.971	0.827
	Shuffled O2	0.664	0.819	0.627

Table A.7: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.732	0.866	0.753
	Soft O2	0.976	0.988	0.869
	Hard hydr.	1.000	1.000	0.897
	Shuffled O2	0.708	0.854	0.705
advice-recall	FlatIP	0.444	0.559	0.483
	Soft O2	0.652	0.662	0.582
	Hard hydr.	0.674	0.674	0.601
	Shuffled O2	0.424	0.549	0.454
u-param	FlatIP	0.739	0.853	0.744
	Soft O2	0.946	0.962	0.844
	Hard hydr.	0.980	0.980	0.864
	Shuffled O2	0.729	0.845	0.705

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	Channel	AH@10
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460